

EDUCATION

- **Ph.D., University of Northern British Columbia** **Prince George, Canada**
Natural Resources & Environmental Studies *Sep.2021 – present*
Supervisor: Ken Otter
- **Master of Science, University of British Columbia** **Vancouver, Canada**
Department of Forest Resources Management *Sep.2017 – Nov.2019*
Supervisor: Bianca N.I. Eskelson
- **Bachelor of Science, National Taiwan University** **Taipei, Taiwan**
School of Forestry and Resource Conservation & Department of Physics *Sep.2012 – Jun.2017*
Double Major Student
Supervisor: Biing T. Guan, Tzeng-Yih Lam, and Shi-Wei Chu

PUBLICATION

JOURNAL ARTICLE

- **Tseng, S.** and S. Kahl. **2025** birdnetTools: An R package for working with BirdNET output. *Journal of Open Source Software*. (Submitted)
- Baum, J.K., Slein, M.A., Garen, J.C., . . . , **Tseng, S.**, et al. **2025** Widespread ecological responses and cascading effects of the record-breaking 2021 North American heatwave. *Nature Ecology and Evolution*. (In Revision)
- Darras, K., Rountree, R., Van Wilgenburg, S., . . . , **Tseng, S.**, et al. **2025** Worldwide Soundscapes: a synthesis of passive acoustic monitoring across realms. *Global Ecology and Biogeography* ([link](#))
- **Tseng, S.**, Hodder, D., and Otter, K. **2025** Setting BirdNET confidence thresholds: Species-specific vs. universal approaches. *Journal of Ornithology* ([link](#))
- **Tseng, S.**, Hodder, D., and Otter, K. **2024** Using Autonomous Recording Units for Vocal Individuality: Insights from Barred Owl Identification. *Avian Conservation and Ecology* ([link](#))
- Pelletier, F., Eskelson, B.N.I., Monleon, V.J., and **Tseng, Y.-C.** **2021** Using Landsat imagery to assess burn severity of National Forest Inventory plots. *Remote Sensing* ([link](#))
- **Tseng, Y.-C.**, Eskelson, B.N.I., Martin, K., and LeMay, V. **2021**. Automatic bird sound detection: Logistic model based acoustic occupancy model. *Bioacoustics* ([link](#))
- **Tseng, Y.-C.**, and Chu, S.-W. **2017**. High spatio-temporal-resolution detection of chlorophyll fluorescence dynamics from a single chloroplast with confocal imaging fluorometer. *Plant Methods* ([link](#))

OPEN-SOURCE SOFTWARE PACKAGES

- **birdnetTools** (2025) — R package for post-processing, visualization, and validation of BirdNET output. Developed in collaboration with the BirdNET team. Available on GitHub. [Documentation](#)
- **bbsTaiwan** (2024) — Tools for accessing and analyzing Taiwan Breeding Bird Survey (BBS) data. Developed for national biodiversity research and public engagement. [Documentation](#)

POPULAR SCIENCE ARTICLE.....

- **2024** Why do we band birds at all? *Nature Conservation Quarterly*, 124. ([link](#))
- **2023** Nest boxes for barn owl conservation. *Nature Conservation Quarterly*, 122. ([link](#))
- **2023** When beaver dam collapses. *Nature Conservation Quarterly*, 121. ([link](#))
- **2023** Listening to the whispers of the wings. *Discovery*, 50(1):85-90. ([link](#))
- **2022** Hey, there are wild black bears in my backyard. *Nature Conservation Quarterly*, 117. ([link](#))

CONFERENCE TALK / POSTER ([check full list here](#)).....

- **SCO-SOC Annual Meeting** **Saskatoon, Canada**
BirdNET workshop organizer, invited seminar speaker Aug.2025
- **Ecological Data Science Summit** **Santa Barbara, USA**
Invited by conference organizer for discussion of current AI use in ecology Feb.2025
- **Posit::conf** **Seattle, USA**
Selected as one of the 40 opportunity scholars, early career software developer Aug.2024
- **AFO SCO-SOC WOS Joint Meeting** **Peoria, USA**
Evaluation and application of BirdNET: identification of bird species by their sounds (Oral) Aug.2024
- **International Ecoacoustic Congress** **Madrid, Spain**
Setting BirdNET confidence thresholds: Species-specific vs. universal approaches (Oral) Jul.2024
- **Congress of Animal Behavior and Ecology** **Taipei, Taiwan**
Avian biodiversity across wide spatiotemporal scale using passive acoustic monitoring (Oral) Feb.2024

RECOGNITION

SCHOLARSHIP and RESEARCH GRANT ([check full list here](#)).....

- **Bert Brink scholarship** **BC Nature**
Outstanding graduate student for natural resources research in BC Aug.2025
- **Michael Smith Foreign Study Supplements** **NSERC**
Developing an R package for BirdNET in Chemnitz, Germany June.2025
- **Early Career Grant** **National Geographic**
Leading a fieldwork in Lithuania for avian acoustic monitoring May.2025
- **Hugh Hamilton Memorial Scholarship** **Nature Vancouver**
For graduate student with outstanding academic & involvement with Nature Science. May.2024
- **rOpenSci Championship Program** **rOpenSci**
Outstanding proposal for R package development. Jan.2024
- **Canada Graduate Scholarship - Doctorial (CGS-D)** **NSERC**
Rewarding and retaining high-calibre doctoral students at Canadian institutions. Apr.2023

AWARD ([check full list here](#)).....

- **SCO-SOC Taverner Award** **SCO-SOC**
Awarded for outstanding ornithological research in Canada Apr.2025
- **Student Presentation Award** **SCO-SOC**
Best oral presentation in the annual conference in Peoria, Illinois Aug.2024
- **Best Student Presentation Award** **BC & CAN TWS**
Best student oral presentation in The Wildlife Society conference Mar.2023